

Impact of Dropping deposit_type

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Initial Model Analysis

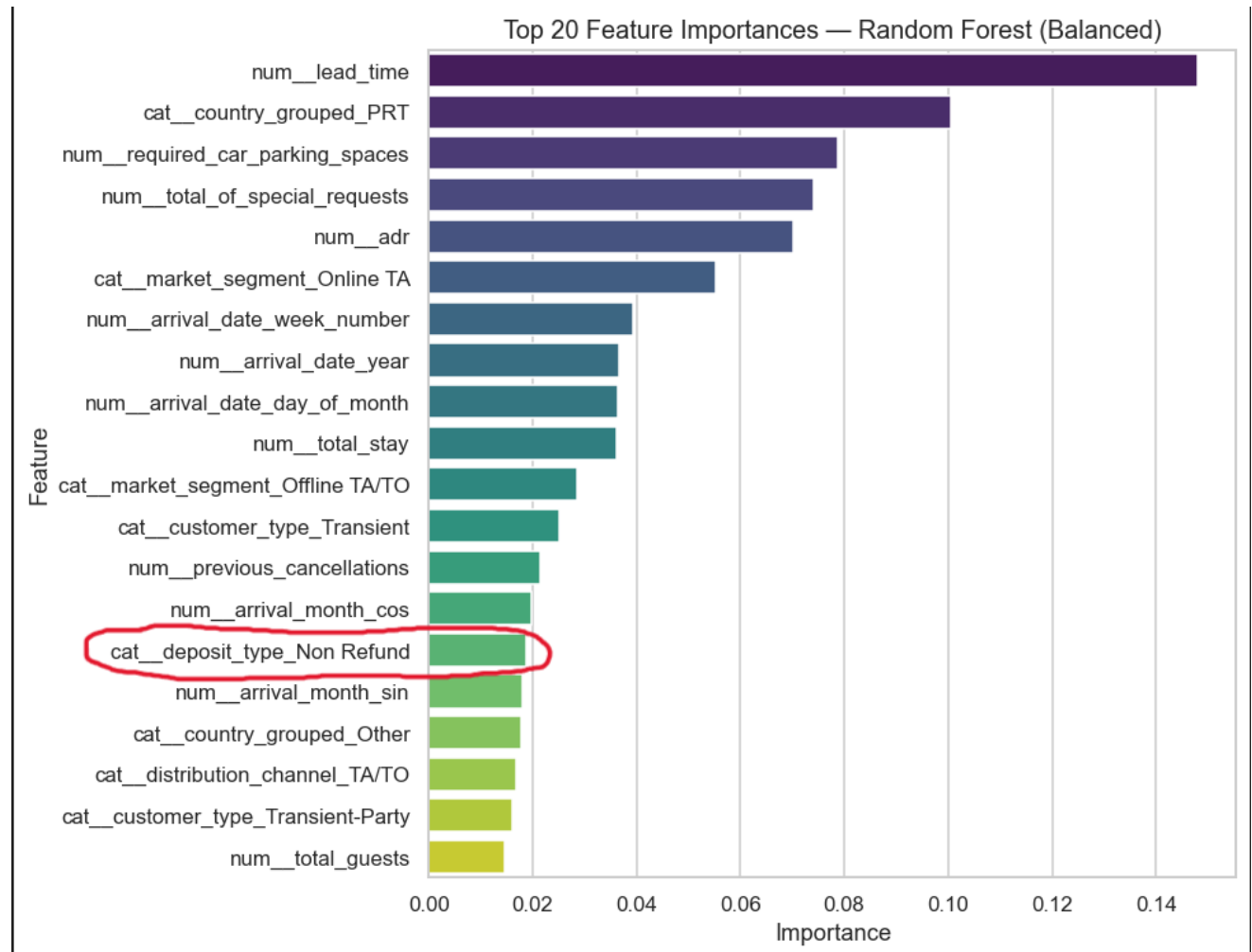
Initial Model Analysis

1a. Model Comparison Table & Model CV Table

	Model	Train Acc	Test Acc	Recall	F1	Balanced Accuracy	ROC-AUC	Overfit Gap
0	Random Forest (Balanced)	0.866833	0.815878	0.802125	0.705699	0.811613	0.898895	0.0510
1	Bagging (Tuned)	0.836622	0.806993	0.826078	0.702009	0.812912	0.899217	0.0296
2	Bagging	0.994052	0.840183	0.645491	0.689740	0.779800	0.899747	0.1539
3	Random Forest	0.973731	0.841502	0.625703	0.684828	0.774572	0.903422	0.1322
4	Gradient Boosting	0.859510	0.836859	0.632368	0.680870	0.773437	0.899350	0.0227
5	Gradient Boosting (Tuned)	0.850409	0.836400	0.628411	0.678893	0.771893	0.899613	0.0140
6	Decision Tree (Balanced)	0.808275	0.772944	0.838784	0.670329	0.793364	0.857182	0.0353
7	Decision Tree	0.822004	0.819547	0.602375	0.647559	0.752192	0.876879	0.0025
8	Ridge (Poly + Balanced)	0.756395	0.757294	0.779004	0.638552	0.764028	0.849155	-0.0009
9	Logistic (Poly + Balanced)	0.756295	0.757008	0.779629	0.638465	0.764024	0.849167	-0.0007
10	KNN (Tuned)	0.996575	0.815707	0.582170	0.634867	0.743276	0.858266	0.1809
11	AdaBoost (Tuned)	0.775900	0.782631	0.670277	0.629253	0.747785	0.817589	-0.0067
12	KNN	0.836680	0.811178	0.561341	0.620682	0.733692	0.851678	0.0255
13	Ridge (Balanced)	0.738725	0.738836	0.774839	0.620207	0.750002	0.834277	-0.0001
14	Logistic (Balanced)	0.738725	0.738836	0.774839	0.620207	0.750002	0.834277	-0.0001
15	Lasso (Balanced)	0.738968	0.738550	0.774630	0.619885	0.749740	0.834367	0.0004
16	AdaBoost	0.793872	0.801261	0.476776	0.569049	0.700623	0.846614	-0.0074
17	Logistic Regression	0.783166	0.788077	0.441366	0.534089	0.680546	0.833799	-0.0049
18	Lasso (L1)	0.783209	0.788019	0.440950	0.533787	0.680377	0.833921	-0.0048
19	Ridge (L2)	0.783023	0.787905	0.441158	0.533770	0.680362	0.833819	-0.0049

	Model	Best CV F1	Type
0	Random Forest (Balanced)	0.697096	Improved
1	Bagging (Tuned)	0.691534	Improved
2	Bagging	0.678892	Original
3	Random Forest	0.678382	Original
4	Gradient Boosting	0.671666	Original
5	Gradient Boosting (Tuned)	0.670809	Improved
6	Decision Tree (Balanced)	0.655674	Improved
7	Decision Tree	0.636967	Original
8	Logistic (Poly + Balanced)	0.634795	Improved
9	Ridge (Poly + Balanced)	0.634712	Improved
10	Lasso (Balanced)	0.620041	Improved
11	Ridge (Balanced)	0.619951	Improved
12	Logistic (Balanced)	0.619895	Improved
13	KNN (Tuned)	0.617683	Improved
14	KNN	0.602368	Original
15	AdaBoost (Tuned)	0.594757	Improved
16	AdaBoost	0.545746	Original
17	Lasso (L1)	0.522243	Original
18	Ridge (L2)	0.521969	Original
19	Logistic Regression	0.521545	Original

1b. Best Model Insights

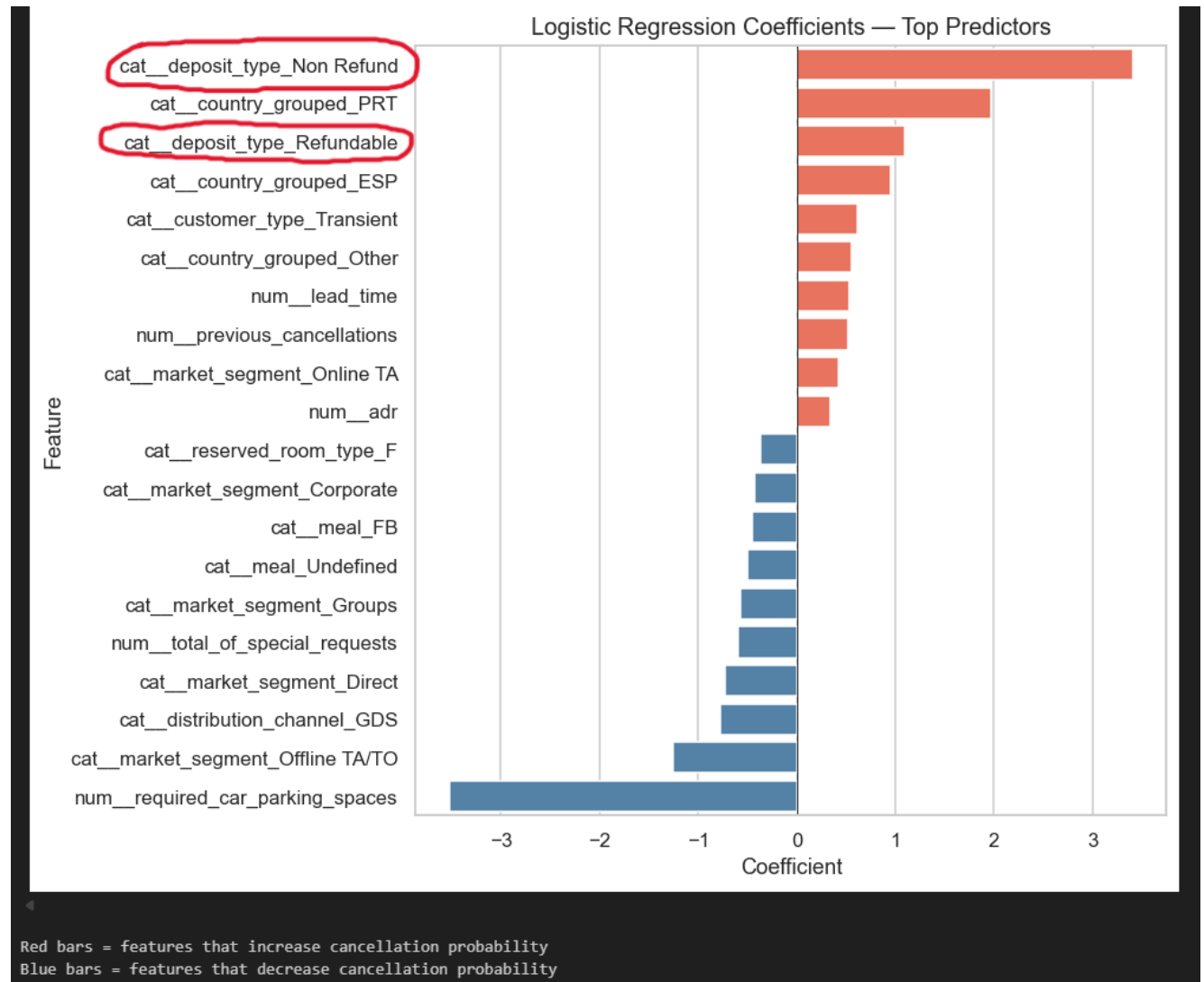


Feature Importance - Tuned Random Forest

Total features after encoding: 50

Non-Refundable deposits are an important feature in our tuned Random Forest model (best model).

1c. Logistic Regression Coefficients



Logistic Regression Top Predictors - Coefficients

Non-Refundable & Refundable deposits have really high coefficients for logistic regression.

New Model (deposit_type dropped) Analysis

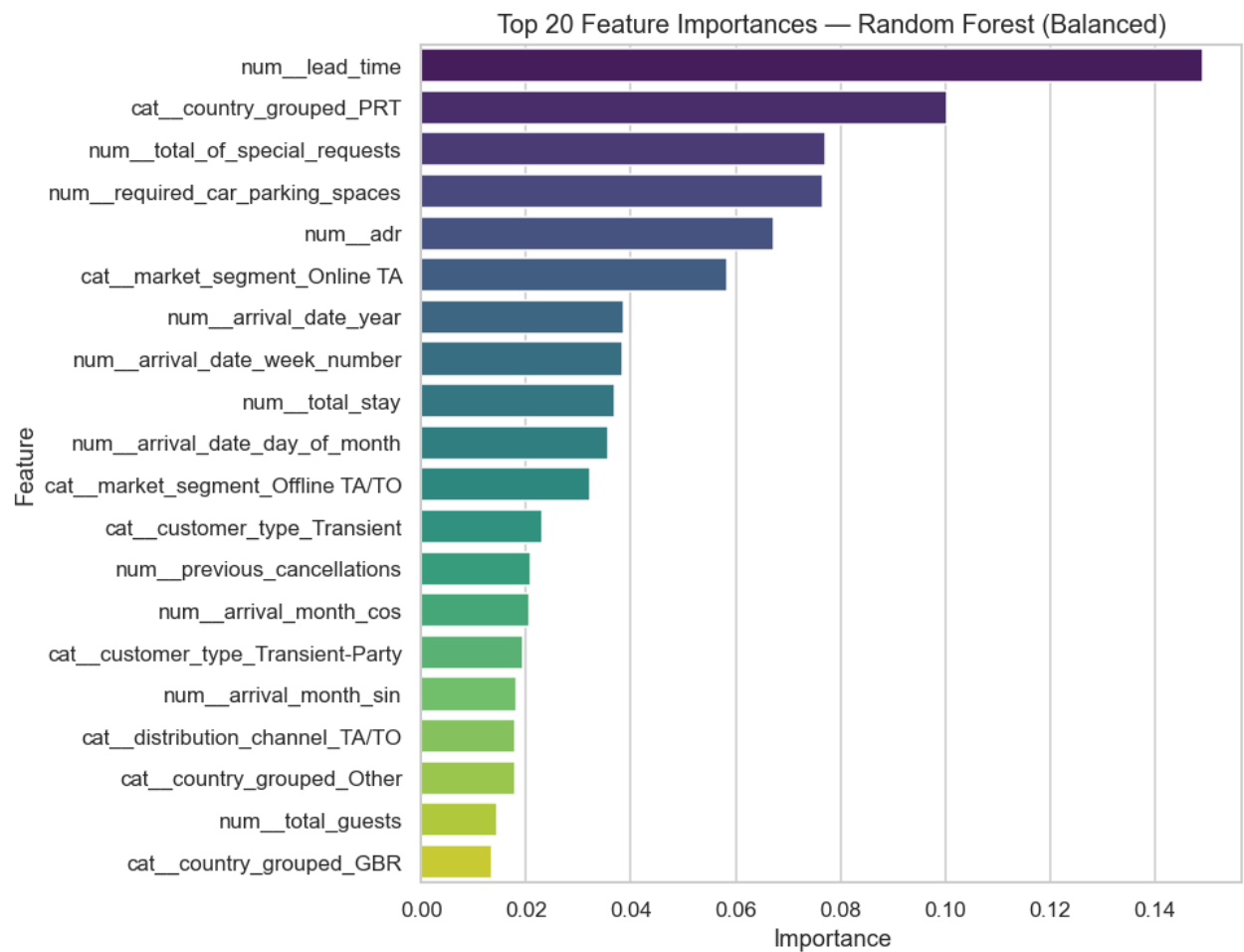
New Model (dropped deposit_type) Analysis

2a. Model Comparison Table & Model CV Table

	Model	Train Acc	Test Acc	Recall	F1	Balanced Accuracy	ROC-AUC	Overfit Gap
0	Random Forest (Balanced)	0.859553	0.811694	0.807957	0.702526	0.810535	0.896856	0.0479
1	Bagging (Tuned)	0.837984	0.808828	0.815247	0.701245	0.810819	0.897943	0.0292
2	Bagging	0.993966	0.839266	0.646115	0.688721	0.779361	0.898898	0.1547
3	Random Forest	0.996560	0.841215	0.632785	0.686864	0.776571	0.899376	0.1553
4	Gradient Boosting	0.859338	0.836973	0.636742	0.682518	0.774872	0.899813	0.0224
5	Gradient Boosting (Tuned)	0.851011	0.835942	0.632785	0.679794	0.772933	0.898336	0.0151
6	Decision Tree (Balanced)	0.810554	0.773116	0.836284	0.669837	0.792707	0.855935	0.0374
7	Decision Tree	0.820915	0.819776	0.605707	0.649107	0.753383	0.875438	0.0011
8	Ridge (Poly + Balanced)	0.755593	0.757008	0.784420	0.639878	0.765510	0.848054	-0.0014
9	Logistic (Poly + Balanced)	0.755535	0.756950	0.784420	0.639823	0.765470	0.848067	-0.0014
10	KNN (Tuned)	0.996560	0.815191	0.582379	0.634301	0.742985	0.857965	0.1814
11	AdaBoost (Tuned)	0.775900	0.782631	0.670277	0.629253	0.747785	0.817589	-0.0067
12	Logistic (Balanced)	0.737034	0.738722	0.780046	0.621680	0.751538	0.832909	-0.0017
13	Lasso (Balanced)	0.736890	0.738091	0.780879	0.621364	0.751362	0.833018	-0.0012
14	Ridge (Balanced)	0.736905	0.738034	0.780046	0.621061	0.751064	0.832887	-0.0011
15	KNN	0.836565	0.810891	0.562383	0.620761	0.733817	0.851079	0.0257
16	Logistic Regression	0.780787	0.786758	0.444074	0.534068	0.680476	0.832584	-0.0060
17	Lasso (L1)	0.780959	0.786758	0.443658	0.533835	0.680347	0.832705	-0.0058
18	Ridge (L2)	0.780744	0.786586	0.443033	0.533283	0.680034	0.832508	-0.0058
19	AdaBoost	0.783711	0.792204	0.427619	0.531109	0.679129	0.843535	-0.0085

	Model	Best CV F1	Type
0	Random Forest (Balanced)	0.694267	Improved
1	Bagging (Tuned)	0.692424	Improved
2	Bagging	0.679146	Original
3	Random Forest	0.676011	Original
4	Gradient Boosting	0.671382	Original
5	Gradient Boosting (Tuned)	0.671334	Improved
6	Decision Tree (Balanced)	0.654153	Improved
7	Decision Tree	0.635569	Original
8	Ridge (Poly + Balanced)	0.634368	Improved
9	Logistic (Poly + Balanced)	0.634203	Improved
10	Ridge (Balanced)	0.619701	Improved
11	Lasso (Balanced)	0.619454	Improved
12	Logistic (Balanced)	0.619291	Improved
13	KNN (Tuned)	0.617524	Improved
14	KNN	0.601564	Original
15	AdaBoost (Tuned)	0.596996	Improved
16	AdaBoost	0.540903	Original
17	Logistic Regression	0.519626	Original
18	Ridge (L2)	0.519623	Original
19	Lasso (L1)	0.519526	Original

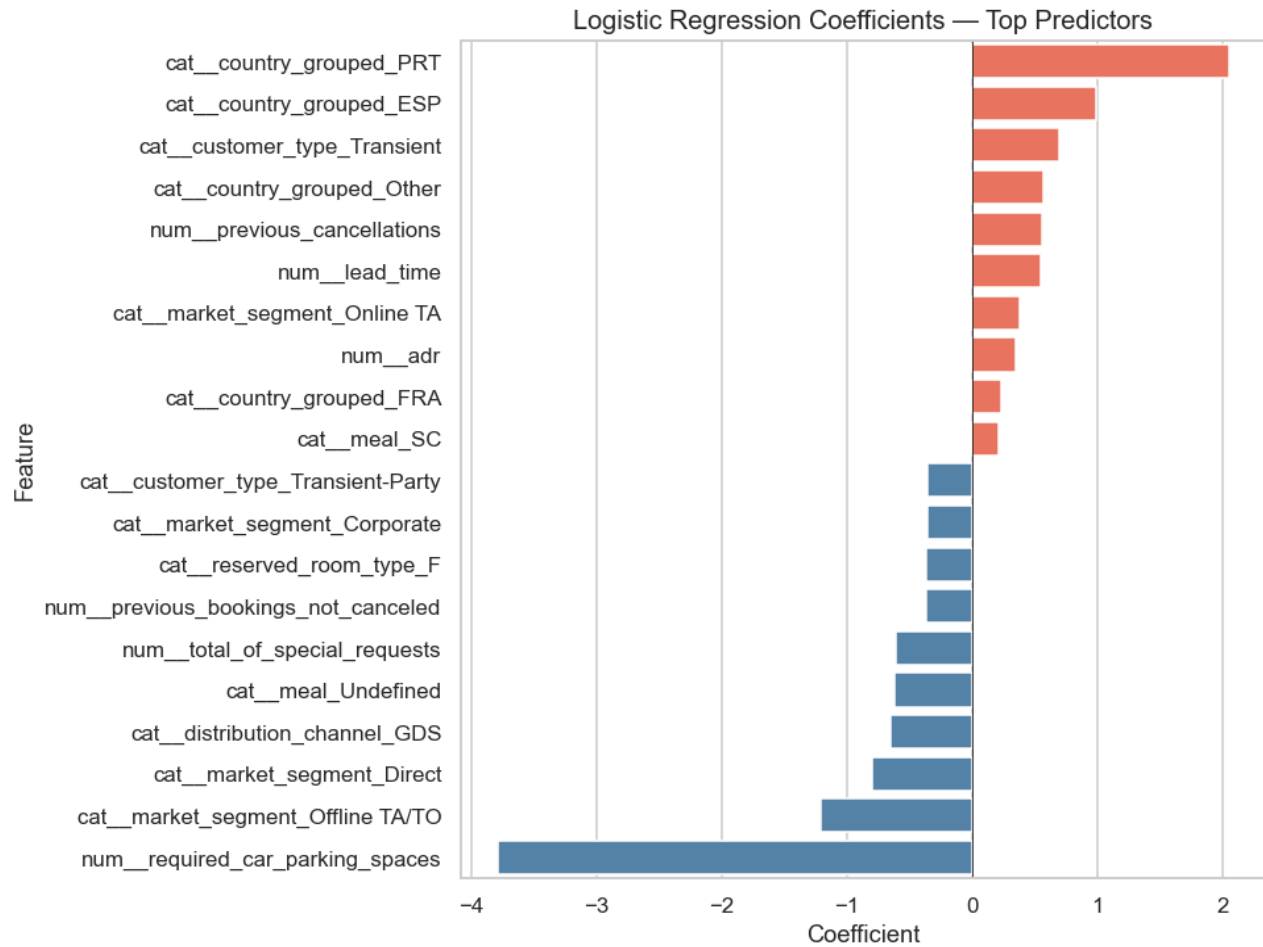
2b. Best Model Insights



Feature Importance - Tuned Random Forest

Total features after encoding: 48

2c. Logistic Regression Coefficients

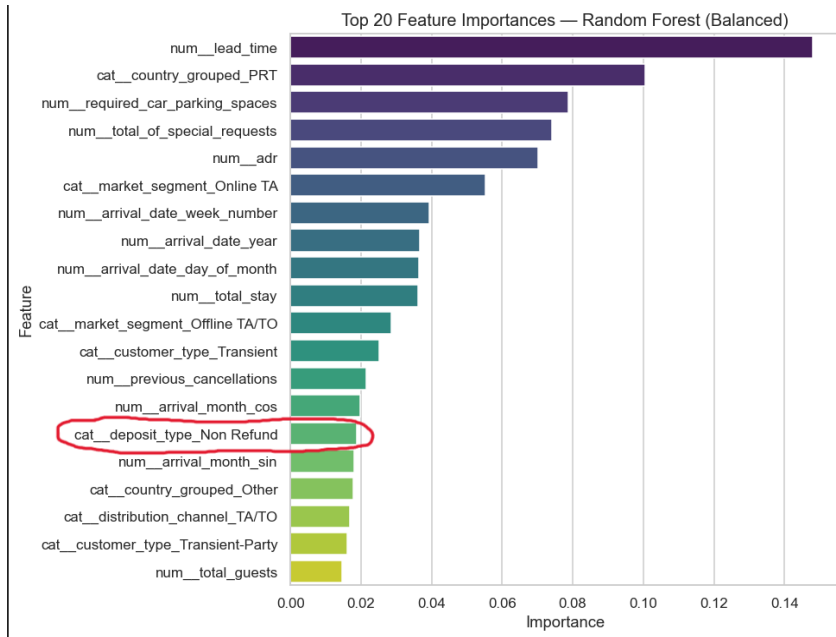


Logistic Regression Top Predictors - Coefficients

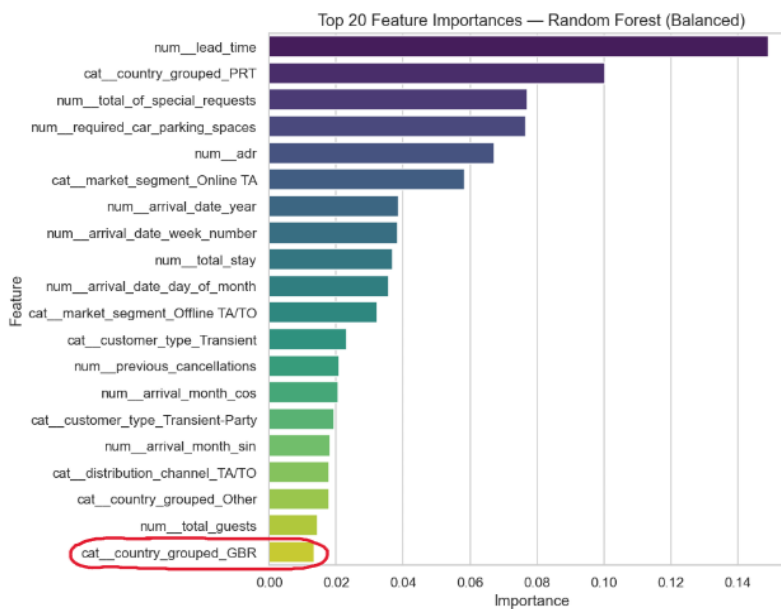
Comparison

Comparison

3a. Feature Importance (Tuned Random Forest) Comparison



^ Initial Feature Importance - Tuned Random Forest (Total Features: 50)



^ New (*deposit_type* dropped) Feature Importance - Tuned Random Forest (Total Features: 48)

	Before (With deposit_type)	After (dropped deposit_type)
Top Performer	num_lead_time	num_lead_time
Removed	cat_deposit_type_Non Refund	–
New Selection	–	cat_country_grouped_GBR
Rank Shifts	<u>Original Ranking:</u> <u>*1 & 2 same*</u> 3. num_required_car_parking_spaces 4. num_total_of_special_requests <u>5 & 6 same</u> 7. num_arrival_date_week_number 8. num_arrival_date_year 9. num_arrival_date_day_of_month 10. num_total_stay <u>*11 - 14 same*</u> 15. cat_deposit_type_Non Refund <u>*16 same*</u> 17. cat_country_grouped_Other 18. cat_distribution_channel_TA/TO 19. cat_customer_type_Transient-Party 20. num_total_guests	<u>New Ranking:</u> <u>*1 & 2 same*</u> 3. num_total_of_special_requests 4. num_required_car_parking_spaces <u>*5 & 6 same*</u> 7. num_arrival_date_year 8. num_arrival_date_week_number 9. num_total_stay 10. num_arrival_date_day_of_month <u>*11 - 14 same*</u> 15. cat_customer_type_Transient-Party <u>*16 same*</u> 17. cat_distribution_channel_TA/TO 18. cat_country_grouped_Other 19. num_total_guests 20. cat_country_grouped_GBR

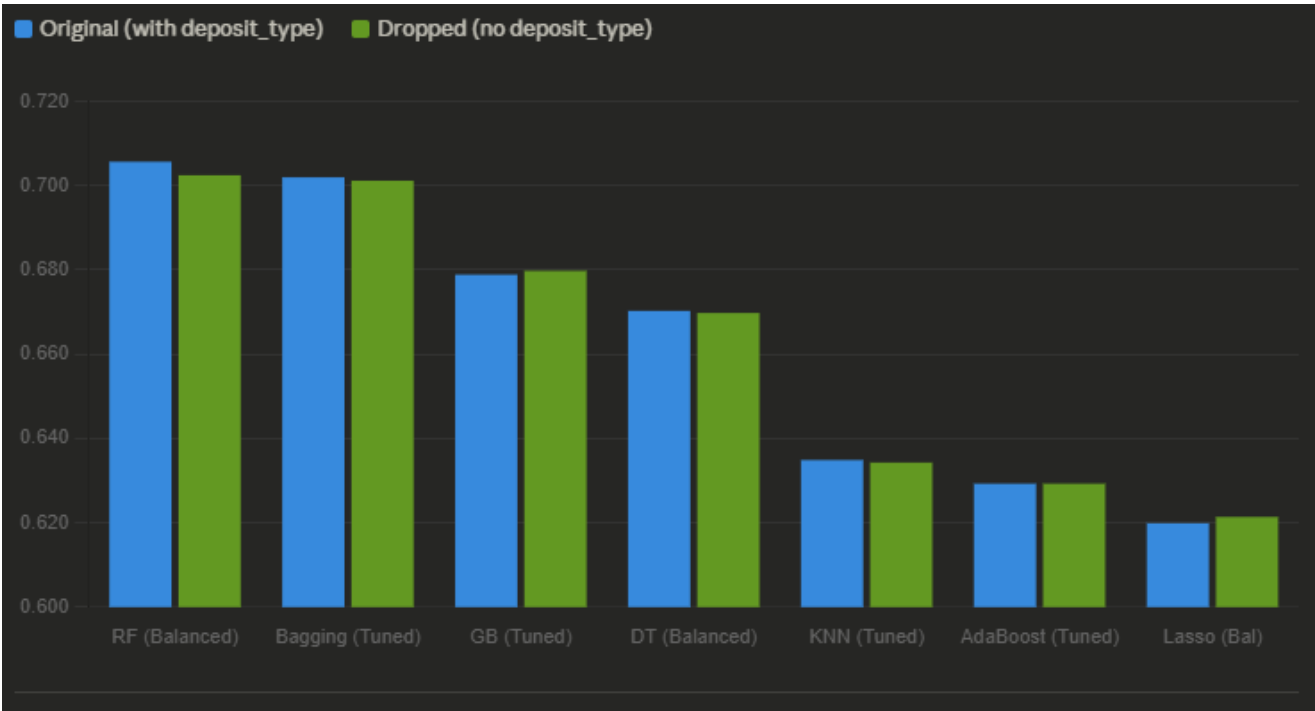
- **Big Three:** Lead Time, Country (Portugal), Special Requests (still dominant predictors)
- **After removing deposit_type:** Great Britain emerged as a new significant predictor
- **Top 20 almost the same:** with minor shifts in ranking and the addition of one predictor
- **Efficiency & focus:** now operates with 48 features (down from 50) without losing predictive power

3b. Key Metrics - Best Model (Tuned Random Forest)

Metric	Test F1	Recall	Test	ROC-AUC
New Model	0.7025	80.8%	81.2%	0.8969
Initial Model	0.7057	80.2%	81.6%	0.8989
Difference	↓ 0.0032	↑ 0.6%	↓ 0.4%	↓ 0.0020

3c. Model Comparison - Original vs Dropped

F1 Model Comparison



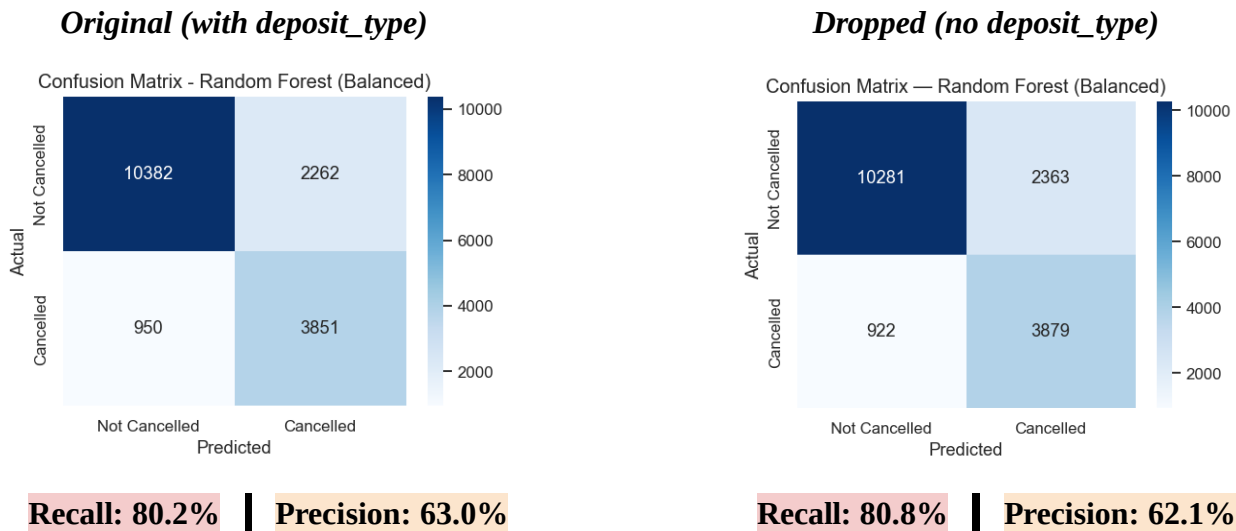
Full Model Comparison

Model	Test F1 (original)	Test F1 (dropped)	Change	CV F1 (original)	CV F1 (dropped)
(Tuned) Random Forest	0.7057	0.7025	↓ 0.0032	0.6971	0.6943
(Tuned) Bagging	0.7020	0.7012	↓ 0.0008	0.6915	0.6924

(Tuned) Gradient Boosting	0.6789	0.6798	↑ 0.0009	0.6708	0.6713
(Tuned) Decision Tree	0.6703	0.6698	↓ 0.0005	0.6557	0.6542
(Tuned) kNN	0.6349	0.6343	↓ 0.0006	—	—
(Tuned) AdaBoost	0.6293	0.6293	=	0.5948	0.5970
(Tuned) Lasso	0.6199	0.6214	↑ 0.0015	—	—



3d. Confusion Matrix - Best Model (Tuned Random Forest) Comparison



3e. Threshold Optimization | (Tuned Gradient Boosting) Comparison

Original Default Threshold: 0.5 | F1: 0.6789 | 3,017 cancellations caught

Dropped Default Threshold: 0.5 | F1: 0.6798 | 3,038 cancellations caught

Optimal Threshold (max F1)	Cancellations Caught
Original: 0.3439 F1: 0.7099 +0.031 vs default	Original: 3,736 +719 more cancellations vs default
Dropped: 0.3297	Dropped: 3,791

F1: 0.7082 +0.028 vs default	+753 more cancellations vs default
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3f. Takeaways/Conclusion

Bottom line: Dropping deposit_type costs only ~ 0.003 F1 on ensemble models. This is a negligible difference, while the leakage concern it eliminates is real. The models remain highly competitive without the deposit_type predictor. The results strongly support keeping deposit_type dropped.

- **Performance drop is negligible:** removing deposit_type costs about 0.003 F1 on top models like random forest and bagging. That's well within noise range and not a meaningful degradation. Our models are essentially just as good without this predictor.
- **Leakage concern is real:** deposit_type_Non Refund was picked by forward selection originally (for both Logistic and Ridge). A feature showing almost 100% cancellation rate is almost certainly capturing outcome information rather than a genuine prebooking signal. Dropping it makes models substantially more defensible for real deployment.
- **Feature selection adapted cleanly:** When deposit_type was removed, forward selection didn't struggle, it simply swapped in legitimate alternatives like arrival_date_year and country_grouped_FRA. No collapse in the selected feature quality.
(worth noting: adr was originally in both logistic and ridge forward selection list, when deposit_type was dropped, forward selection replaced both deposit_type and adr)
- **Recall improved slightly:** Dropped version catches slightly more cancellations at the best model (Random Forest: 80.8% vs 80.2%), which is the metric hotels care most about. This is counterintuitive but real, it suggests deposit_type may have been nudging the model toward confident predictions rather than genuinely improving boundary detection.
- **Gradient Boosting and Lasso both improved slightly:** This is interesting, for models less reliant on that one dominant feature, the cleaner feature set helped marginally.

Models performed comparably without the leaky feature. In a real hotel system, deposit_type at booking time would likely encode policy choices that vary by property (making it unreliable as a generalizable predictor anyways). Dropping deposit_type makes our models safer and more honest.

Diving Deeper

Diving Deeper

4a. deposit_type vs Raw Dataset

- 119,390 Observations, 32 Predictors
- Before data cleaning, so still has duplicate rows, invalid rows, and outliers

Deposit Type	Proportion of Dataset (%)	# of Rows	Cancellation Rate (%)
No Deposit	~ 87.65%	~ 104,641 rows	~ 28.38%
Non Refundable	~ 12.22%	~ 14,587 rows	~ 99.36%
Refundable	~ 0.13%	~ 162 rows	~ 22.22%

No deposits have 28.38% cancellation rate and make up 87.65% of the dataset.

Non Refundable deposits have 99.36% cancellation rate and make up 12.22% of the dataset.

Refundable deposits have 22.22% cancellation rate and make up 0.13% of the dataset.

4b. deposit_type vs Clean Dataset

- 87,222 Observations, 32 Predictors (73.1% of raw dataset remain)
- Dropped duplicate rows, invalid rows, and outliers [32,168 rows (26.9%) dropped]

Deposit Type	Proportion of Dataset (%)	# of Rows	Cancellation Rate (%)
No Deposit	~ 98.69%	~ 86,078 rows	~ 26.72%
Non Refundable	~ 1.19%	~ 1,037 rows	~ 94.70%
Refundable	~ 0.12%	~ 107 rows	~ 24.30%

No deposits now have 26.72% cancellation rate and make up 98.69% of the dataset.

Non Refundable deposits now have 94.70% cancellation rate and make up 1.19% of the dataset.

Refundable deposits now have 24.30% cancellation rate and make up 0.12% of the dataset.

4c. Changes/Interpretation

Deposit Type	Change in Proportion of Dataset (%)	Change in # of Rows	Change in Cancellation Rate (%)
No Deposit	87.65% → 98.69% 11.04% increase	104,641 → 86,078 18,563 rows dropped	28.38% → 26.72% 1.66% decrease
Non Refundable	12.22% → 1.19% 11.03% decrease	14,587 → 1,037 13,550 rows dropped	99.36% → 94.70% 4.66% decrease
Refundable	0.13% → 0.12% 0.01% decrease	162 → 107 55 rows dropped	22.22% → 24.30% 2.08% increase

Deposit Type	Rows Dropped / Original Rows (%)
No Deposit	18,563 / 104,641 = ~ 17.74%
Non Refundable	13,550 / 14,587 = ~ 92.89%
Refundable	55 / 162 = ~ 33.95%

No deposits: 17.74% of rows were either duplicate, invalid, and/or outliers

Non Refundable: 92.89% !!! of rows were either duplicate, invalid, and/or outliers

Refundable deposits: 33.95% of rows were either duplicate, invalid, and/or outliers

4d. Takeaways/Conclusion

Non-Refundable Deposits

- Initially, deposit_type looked strong on the raw data (99.36% cancellation rate for Non Refundable deposits)
- After cleaning the data (dropping duplicates, invalid rows, & outliers), Non Refundable category dropped from 14,587 rows to 1,037 rows

- Meaning 92.89% of Non Refundable bookings were either duplicates, invalid rows, and/or outliers, likely repeated group booking records

No Deposit / Refundable Deposits

- For 98.81% of the cleaned data, deposit_type was either:
 - “No Deposit” (98.69% of dataset, with 26.72% cancellation rate)
 - OR
 - “Refundable” (0.12% of dataset, with 24.3% cancellation rate)
- Both are close to the baseline cancellation rate (27.52%)
- Essentially, deposit_type was adding almost no signal for the vast majority of rows. The models were already learning cancellation patterns from other predictors (e.g. lead_time, total_of_special_requests, etc)
- The reason models didn’t change much after dropping deposit_type was because they only lost predictive power for the 1,037 rows where cancellation rate was 94.70%

Before Dropping

Before Dropping 'deposit_type'

Rank	Feature (<u>Logistic Forward</u>)	Coef	Feature (<u>Logistic Backward</u>)	Coef
1	cat_deposit_type_Non Refund	3.5162	num_required_car_parking_spaces	-6.3956
2	num_required_car_parking_spaces	-2.6316	cat_market_segment_Online TA	1.9428
3	cat_market_segment_Online TA	1.7156	cat_country_grouped_PRT	1.6081
4	cat_country_grouped_PRT	1.4175	num_previous_cancellations	0.6884
5	cat_customer_type_Transient-Party	-0.7553	num_previous_bookings_not_canceled	-0.6188
6	num_previous_cancellations	0.5568	cat_country_grouped_ESP	0.5634
7	num_total_of_special_requests	-0.5502	num_total_of_special_requests	-0.5583
8	num_lead_time	0.4987	num_lead_time	0.4127
9	num_previous_bookings_not_canceled	-0.4772	num_arrival_date_year	0.2870
10	num_adr	0.3243	num_total_stay	0.1772

Both Logistic Regression & Ridge chose: 'num_required_car_parking_spaces', 'cat_market_segment_Online TA', 'num_previous_cancellations', 'num_total_of_special_requests', 'num_previous_bookings_not_canceled', 'cat_country_grouped_PRT', 'num_lead_time' (7 total)

Rank	Feature (<u>Ridge Forward</u>)	Coef	Feature (<u>Ridge Backward</u>)	Coef
1	cat_deposit_type_Non Refund	3.5162	num_required_car_parking_spaces	-6.3945
2	num_required_car_parking_spaces	-2.6316	cat_market_segment_Online TA	1.6273
3	cat_market_segment_Online TA	1.7156	cat_country_grouped_PRT	1.5726
4	cat_country_grouped_PRT	1.4175	cat_customer_type_Transient	0.8461
5	cat_customer_type_Transient-Party	-0.7553	num_previous_cancellations	0.7068
6	num_previous_cancellations	0.5568	num_previous_bookings_not_canceled	-0.6232
7	num_total_of_special_requests	-0.5502	num_total_of_special_requests	-0.5631
8	num_lead_time	0.4987	num_lead_time	0.5109

9	num_previous_bookings_not_canceled	-0.4772	num_adr	0.2934
10	num_adr	0.3243	num_arrival_date_year	0.1831

- deposit_type_Non Refund dominates both forward selection lists with the largest coefficient (+3.52) ~ roughly double the next feature in forward selection (market_segment_Online TA at +1.72)
- This is a red flag: a single one-hot category shouldn't dwarf every other predictor
- Forward selection chose it first because it's a shortcut: 94.7% of Non Refund bookings cancel
- Backward selection never relied on it: required_car_parking_spaces took the top spot instead (-6.40)
- Forward and backward selection disagree on the most important feature, suggesting deposit_type is distorting feature selection rather than providing genuine signal
- Both methods still agree on 7 core features, but forward lists are built around deposit_type while backward lists find a balanced set without it

After Dropping

After Dropping 'deposit_type'

Rank	Feature (Logistic Forward)	Coef	Feature (Logistic Backward)	Coef
1	num_required_car_parking_spaces	-6.3861	num_required_car_parking_spaces	-6.2698
2	cat_market_segment_Online TA	1.7624	cat_market_segment_Online TA	1.8233
3	cat_country_grouped_PRT	1.4855	cat_country_grouped_PRT	1.5817
4	cat_customer_type_Transient-Party	-0.8295	num_previous_cancellations	0.6789
5	num_previous_cancellations	0.6950	num_total_of_special_requests	-0.5686
6	num_previous_bookings_not_canceled	-0.6797	num_previous_bookings_not_canceled	-0.5494
7	num_total_of_special_requests	-0.5561	num_lead_time	0.4171
8	num_lead_time	0.4821	num_adr	0.3051
9	num_arrival_date_year	0.2529	num_arrival_date_year	0.2335
10	cat_country_grouped_FRA	-0.2179	num_total_stay	0.1550

Both Logistic Regression & Ridge chose: 'num_previous_cancellations', 'num_total_of_special_requests', 'num_lead_time', 'num_arrival_date_year', 'num_required_car_parking_spaces', 'cat_country_grouped_PRT', 'num_previous_bookings_not_canceled', 'cat_market_segment_Online TA' (8 total)

Rank	Feature (Ridge Forward)	Coef	Feature (Ridge Backward)	Coef
1	num_required_car_parking_spaces	-6.3689	num_required_car_parking_spaces	-6.3502
2	cat_market_segment_Online TA	1.7624	cat_market_segment_Online TA	1.8245
3	cat_country_grouped_PRT	1.4855	cat_country_grouped_PRT	1.5824
4	cat_customer_type_Transient-Party	-0.8295	num_previous_cancellations	0.6784
5	num_previous_cancellations	0.6949	num_total_of_special_requests	-0.5691
6	num_previous_bookings_not_canceled	-0.6796	num_previous_bookings_not_canceled	-0.5493
7	num_total_of_special_requests	-0.5561	num_lead_time	0.4170
8	num_lead_time	0.4821	num_adr	0.3062

9	num_arrival_date_year	0.2529	num_arrival_date_year	0.2333
10	cat_country_grouped_FRA	-0.2178	num_total_stay	0.1546

- Forward and backward selection now converge much more cleanly
- Both methods agree on 8 features (up from 7): stronger consensus without the distorting feature
- required_car_parking_spaces is now the dominant feature in all four lists (-6.27 to -6.39), which makes intuitive sense, guests requesting parking signal genuine commitment
- No artificially inflated top feature: rankings follow a smooth gradient from parking through market segment, country, and behavioral features
- Forward selection replaced deposit_type and adr with arrival_date_year and country_grouped_FRA
 - arrival_date_year - it's known at the time of booking, it captures any YoY shifts in cancellation patterns; backward selection already chose it in both LR and Ridge even when deposit_type was still in the dataset, so it's not a replacement for deposit_type, it was already a proven predictor that forward selection now has room to include
 - country_grouped_FRA - ESP and the 'other' category are too close to the baseline (27.5%) to be useful once other features are selected in forward selection; FRA is chosen because it provides a cleaner split that the existing features weren't already covering
- Overall feature selection is more stable and trustworthy

Final Conclusion - Drop
deposit_type?

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Drop deposit_type

Every analysis in this document points in the same direction:

- **Performance impact is negligible:** 0.003 F1 on the best model, well within noise
- **Recall actually improved slightly:** 80.2% → 80.8%, the metric hotels care most about
- **Feature selection became more stable:** forward and backward agree on 8 core features (up from 7)
- **The reason the drop barely mattered is structural:**
 - After cleaning, Non Refund is only 1.19% of the data (1,037 rows)
 - 92.89% of original Non Refund rows were duplicates, invalid, or outliers, the feature's apparent strength was inflated by dirty data
- For 98.81% of the cleaned data, deposit_type carried virtually no signal (both No Deposit and Refundable cancel near the 27.5% baseline)
- Models were already learning from lead_time, special requests, parking, and country; deposit_type was a crutch for ~1,000 rows, not a genuine predictor
- Dropping it produces a cleaner, more defensible model that a hotel could deploy with confidence